

Highlights

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CPAI: A Controlled-Point Discriminant Analysis for IoT Network Intrusion Detection

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ABSTRACT

The proliferation of Internet of Things (IoT) devices poses significant security challenges, particularly intrusion detection, due to evolving cyber attacks and their resource constraints. While existing distance-based approaches like Null Foley-Sammon Transform (NFST) show promise, they suffer from singularity and small sample size problems, limiting their effectiveness in diverse IoT scenarios. In this paper, we introduce Controlled-Point Discriminant Analysis (CPAI), a novel algorithm designed for robust IoT network intrusion detection. CPAI addresses these limitations by effectively controlling the location of projected data points in the discriminant space and employing kernel tricks with dual training strategies - One-vs-All (OvA) for computational efficiency and **One-vs-Rest (OvR)** for enhanced accuracy. Extensive evaluation across five benchmark IoT network security datasets demonstrates CPAI's superiority in terms of accuracy and stability compared to KNFST and other distance-based algorithms. **The CPAI-OvR variant achieves over 98% accuracy on most datasets, with a minimum of 94.64% on the most challenging scenarios, while consistently maintaining false positive rates below 3.35%.**

1. Introduction

The Internet of Things (IoT) has revolutionized various critical sectors (e.g., healthcare, logistics, and manufacturing) by enabling seamless communication and automation over the Internet. While this significantly improves the efficiency of operations and services, it also introduces cybersecurity threats by exposing networks to unauthorized access, data breaches, and service disruptions [2]. Furthermore, the inherent IoT devices often have limited computational and energy resources, making them unsuitable for computationally intensive security mechanisms. Therefore, developing lightweight yet effective security solutions specifically designed for IoT systems is a critical necessity [16].

1.1. Paradigms in Anomaly-Based Intrusion Detection

Anomaly-based Network Intrusion Detection Systems (aNIDS) represent a promising solution for this challenge. Their primary strength lies in their adaptability to automatically identify threats by analyzing deviations from normal network behavior, making them more flexible than signature-based systems that rely on predefined attack patterns. Numerous studies have demonstrated the superior performance of aNIDS in conventional networks [2, 17, 21]. Within the domain of aNIDS, several methodological paradigms have been dominant.

Deep learning (DL) and tree-based algorithms are widely used in proposed aNIDSs due to their effectiveness in processing tabular network traffic data [2, 1, 5]. DL models excel at automatically capturing complex patterns [23]. However, they typically demand substantial computational resources (e.g., GPUs, significant memory), making them impractical for resource-constrained IoT devices. Even though inference can be optimized for edge devices (e.g., via model distillation), the initial training overhead remains a significant barrier. While tree-based models are generally more lightweight, they can struggle in unsupervised or semi-supervised settings common in IoT, where labeled data for novel attacks is limited.

Given these trade-offs, distance-based algorithms have emerged as a promising alternative for IoT network security. These algorithms do not require specialized hardware like GPUs and are highly adaptable in contexts where attack labels are unavailable, making them more suitable for resource-constrained IoT devices [11]. This has led to a renewed interest in methods rooted in Linear Discriminant Analysis (LDA) [12], which offer a powerful middle ground: they provide a principled approach to feature extraction that is both computationally efficient and more interpretable than deep learning "black boxes". Its key concept involves projecting instances from a high-dimensional space into a lower-dimensional subspace through linear transformation, thereby compacting samples and making them easier to classify.

1.2. The Challenge of Null-Space Discriminant Analysis

A state-of-the-art method in this direction is the Null Foley-Sammon Transform (NFST) [15], that extends classical Linear Discriminant Analysis [12, 13]. The theoretical elegance of NFST lies in its objective to project all samples

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of a class to a single, unique point by enforcing zero within-class scatter, thereby maximizing discrimination according to the Fisher criterion. Despite its strong theoretical appeal, our analysis and recent literature reveal two critical, unresolved flaws that severely hinder NFST's reliability in practical IoT scenarios:

- *Singularity problem:* The foundational assumption that projected class points remain distinct is not guaranteed. However, as demonstrated by Ali and Chaudhuri [3], these singularities can erroneously overlap, leading to catastrophic misclassification. The problem gets worse in IoT networks where normal and malicious behaviors could have several similar network characteristics, reducing the model's accuracy. To the best of our knowledge, this singularity problem remains unresolved within the application of aNIDS in IoT networks.
- *Small sample size problem:* The second theoretical need of NFST is the linearity of sample vectors. This assumption yields that the number of samples (n) does not exceed the feature dimensionality (d), i.e., $n \leq d$. Consequently, applying NFST in IoT networks may reduce accuracy or even make it impossible to find a solution.

1.3. Our Contribution: Controlled-Point Discriminant Analysis

Recognizing these critical limitations, we propose a Controlled-Point discriminant Analysis for IoT (CPAI), a novel framework designed to overcome the theoretical drawbacks of NFST while retaining the strengths of discriminant analysis. The core innovation of CPAI is its ability to *explicitly control the placement of projected data points*, a principle that fundamentally resolves the singularity problem. Furthermore, by leveraging the kernel trick, CPAI robustly handles the non-linear, efficiently handling the constraint on the number of samples.

Crucially, the formulation of CPAI as a regularized least-squares problem, it admits a *closed-form analytical solution*. This transforms the learning problem into solving a single system of linear equations. This structural simplicity not only makes the model easier to analyze but also avoids potential issues like local minima and convergence problems inherent in more complex optimizers.

This computational efficiency, combined with its high accuracy, positions CPAI as a framework uniquely aligned with the core requirements of IoT security: *robustness, efficiency, and reliability*. To enhance its flexibility for diverse deployment needs, we propose two training strategies: CPAI-OvA for computational efficiency, and CPAI-OvR for superior accuracy. We conduct extensive experiments on five benchmark IoT datasets, demonstrating that our proposed CPAI framework significantly outperforms not only its predecessor (KNFST) but also a wide range of other strong baseline methods.

The remainder of this paper is organized as follows. Section 3 details the theoretical foundations and implementation of our proposed CPAI framework. A direct comparison between CPAI and its predecessor, NFST, is provided in Appendix B to highlight our key theoretical advancements. Section 5 presents the comprehensive experimental setup, results, and analysis, validating CPAI's superior performance against a wide range of competitors. Finally, Section 6 concludes the paper with a summary of our findings and discusses directions for future work. To ensure the paper is self-contained, an overview of the NFST algorithm is provided in Appendix A.

2. Related Work

Network Intrusion Detection Systems (NIDS) for IoT have been approached from several methodological directions. We survey the principal families of methods, discuss their practical limitations in resource-constrained IoT environments, and identify the gap that motivates the present work.

2.1. IoT Intrusion Detection: Problem Overview

The IoT intrusion detection problem is fundamentally a supervised multi-class classification task: network flows must be assigned to one of several categories, typically a benign class and multiple attack types such as DDoS, botnet, reconnaissance, and data exfiltration [17, 21]. This framing distinguishes it from pure one-class anomaly detection and is the standard adopted by the five benchmark datasets used in this work [18, 4, 20, 22, 30].

IoT environments impose three compounding constraints that significantly restrict the applicability of many IDS methods [16]. First, IoT devices are resource-constrained: limited memory and processing power preclude the deployment of GPU-dependent models at the network edge. Second, IoT traffic is highly heterogeneous, spanning diverse protocols, device types, and attack patterns that vary across deployments. Third, the class imbalance inherent in IoT datasets, where benign traffic vastly outnumbers attack instances, can severely bias classifier performance [2, 17]. A comprehensive taxonomy of IDS approaches for IoT, organized by data source, detection technique, and placement strategy, is provided by Jamalipour and Murali [17]. In the following subsections, we review the major methodological paradigms within this taxonomy and assess their suitability for IoT gateway deployment.

2.2. Deep Learning and Tree-Based Approaches

Deep learning (DL) methods have achieved state-of-the-art detection performance in IoT IDS by automatically learning discriminative feature representations from raw network traffic [1, 2]. Convolutional Neural Networks (CNNs) exploit spatial correlations within traffic feature vectors, while Recurrent Neural Networks (RNNs) — particularly Long Short-Term Memory (LSTM) variants — capture temporal dependencies in sequential traffic flows [17]. Deep Autoencoders (AEs) offer an unsupervised alternative, learning a

compact representation of normal traffic and identifying anomalies through reconstruction error. However, AE-based models face the challenge of representation collapse [25], where the network maps the entire feature manifold to a trivial point during training, losing all discriminative power. Ahmad et al. [1] provide a comprehensive benchmark of DL architectures in IoT IDS, confirming consistently strong detection rates. In the context of DDoS detection specifically, Cil et al. [10] propose a feed-forward DNN that integrates feature extraction and classification within a unified architecture, achieving 99.99% binary detection accuracy and 94.57% multi-class classification accuracy on the CI-CDDoS2019 dataset. A representative example of DL deployment under IoT hardware constraints is RealGuard [23], a lightweight DNN-based NIDS that operates directly on IoT gateways, achieving 99.57% average accuracy while processing over 10,000 packets per second on a Raspberry Pi. Despite these advances, DL methods share a fundamental limitation: their dependence on GPU acceleration and large labeled training sets creates an inherent mismatch with the resource-constrained nature of IoT edge devices [17, 23]. Furthermore, their opaque internal representations reduce interpretability, which is critical in security-sensitive deployments.

Tree-based ensemble methods offer a more resource-efficient alternative. Decision Trees (DTs) provide interpretable, rule-based classification that naturally handles the heterogeneous feature types common in IoT network traffic [17]. Random Forest and XGBoost extend this principle by aggregating predictions from multiple DTs to improve robustness against noise and overfitting. Azam et al. [5] conduct a comparative evaluation of ML-based IDS approaches and demonstrate that tree-based ensembles achieve competitive detection accuracy with substantially lower computational overhead than DL counterparts. Nevertheless, tree-based models are sensitive to feature engineering quality and tend to underperform in semi-supervised settings where labeled data for novel attack classes is limited.

2.3. Distance-Based and Discriminant Analysis Methods

Existing detection methods can be further distinguished by their geometric interpretation of normality. Proximity-based methods such as K-Nearest Neighbor (KNN) [24] and Local Outlier Factor (LOF) [8] identify anomalies by measuring local spatial deviations from known normal behavior, while boundary-based methods including One-Class SVM (OCSVM) [27] and Support Vector Data Description (SVDD) [29] model normality as a minimal enclosing hyperplane or hypersphere. Falcao et al. [11] provide a quantitative comparison of these approaches in the IoT IDS context, confirming their effectiveness for unsupervised anomaly detection. However, proximity-based methods scale poorly with the number of training samples (e.g. KNN requires $\mathcal{O}(n)$ comparisons per inference), while boundary-based methods are sensitive to kernel selection and tend to ignore the intrinsic class structure of the data.

A more principled approach lies in discriminant analysis, which explicitly models the internal variance structure of traffic classes. Fisher's Linear Discriminant Analysis (LDA) [12] provides a theoretically grounded framework for class separability maximization through linear projection, and the Foley-Sammon Transform (FST) [13] extends it by finding a set of optimal, mutually orthogonal discriminant vectors. The Null Foley-Sammon Transform (NFST) [15] represents the state of the art in this direction: by enforcing zero within-class scatter ($\varphi^T S_w \varphi = 0$), NFST aims to collapse all samples of a class to a single, distinct projection point, theoretically maximizing the Fisher criterion in the null space of S_w . Kernelized variants (KNFST) [7, 19] extend this principle to non-linear settings via the kernel trick, enabling application in the $n > d$ regime common in IoT datasets. However, Ali and Chaudhuri [3] demonstrate that NFST's singularity points are not guaranteed to remain distinct: under certain data configurations, different classes collapse onto the same projection coordinate, leading to catastrophic misclassification. This singularity problem is further exacerbated in IoT networks where normal and malicious traffic often share overlapping statistical characteristics. In addition, linear NFST is restricted to the small sample size regime ($n \leq d$), limiting its applicability when the number of training samples exceeds the feature dimensionality. To the best of our knowledge, these two limitations remain unresolved in the prior IoT IDS literature [3, 15]. CPAI is developed to resolve both by replacing the passive singularity discovery of NFST with an explicit, target-driven projection system.

2.4. Federated and Lightweight Approaches

The centralized training paradigm shared by the methods described above raises additional challenges in IoT deployments: aggregating raw network traffic from distributed devices introduces communication overhead and creates data privacy risks, particularly in sensitive industrial and healthcare IoT contexts [16]. Federated Learning (FL) addresses these concerns by enabling collaborative model training across distributed clients without sharing raw data. Buyuktanir et al. [9] provide a comprehensive survey of FL-based IDS solutions across IoT, healthcare, and industrial network domains, demonstrating that FL achieves competitive detection accuracy while substantially improving data privacy. The survey highlights that the integration of Generative Adversarial Networks (GANs) and hybrid DL architectures within FL frameworks represents a promising direction for improving robustness against non-IID data distributions, a persistent challenge in heterogeneous IoT environments. At the intersection of FL and resource efficiency, Bocekci and Yildiz [6] propose a Mutex-based federated architecture for the full fine-tuning of the TinyLlama-1.1B language model on resource-constrained edge clients. While their application domain targets NLP adversarial attack detection rather than network IDS, the proposed sequential GPU-access protocol and FP16 memory optimization offer relevant engineering

Table 1

Representative IoT intrusion detection works organised by methodological category, with evaluation datasets and key strengths and limitations assessed from the perspective of resource-constrained IoT deployment.

Work	Category	Evaluation Dataset(s)	Key Strength	Key Limitation
Ahmad et al. [1]	Deep Learning	BoT-IoT, N-BaloT, others	Comprehensive DL benchmark for IoT IDS; strong detection accuracy	GPU-dependent; high memory footprint
Cil et al. [10]	Deep Learning	CICDDoS2019	Feed-forward DNN achieving 99.99% binary detection accuracy	Single-domain dataset; limited attack diversity
RealGuard [23]	Deep Learning	CICIDS2017	Lightweight DNN deployable on IoT gateways; real-time inference	Requires fully labeled training data; limited to known attacks
Azam et al. [5]	Tree-Based	CICIDS2017, NSL-KDD	Interpretable; computationally efficient ensemble approach	Degrades in semi-supervised and class-imbalanced settings
Falcao et al. [11]	Distance Boundary	/ CICIDS2017, others	Rigorous quantitative comparison of anomaly detectors for IDS	Ignores class structure; poor scalability with n
Guo et al. [15]; Bodesheim et al. [7]	Discriminant Analysis	Face recognition benchmarks	Zero within-class scatter; kernelised variant handles $n > d$	Singularity problem [3]; $n \leq d$ for linear case
Buyuktanir et al. [9]	Federated Learning	IoT, healthcare, industrial	Privacy-preserving; distributed training without raw data sharing	Communication overhead; non-IID data challenges
CPAI (this work)	Discriminant + Kernel	BoT-IoT, IoTID20, ToN-IoT, N-BaloT, CIC-IoT2023	Resolves singularity; closed-form; no GPU; handles $n > d$	Kernel and λ selection required; $\mathcal{O}(n^2)$ training

principles for deploying learning systems under strict hardware constraints - a challenge equally applicable to IoT-edge IDS deployment.

2.5. Summary and Research Gap

Table 1 provides a structured comparison of the surveyed IDS works. The analysis reveals a persistent and unaddressed gap. DL methods achieve high accuracy but require hardware resources incompatible with IoT edge deployment. Tree-based methods are lightweight but require extensive labeled data and struggle in semi-supervised settings. Distance and boundary-based methods are hardware-agnostic but ignore class structure and scale poorly. Discriminant analysis methods, particularly NFST and KNFST, offer the most promising balance of theoretical rigor and efficiency, but are fundamentally compromised by the singularity problem and the $n \leq d$ constraint [3, 15]. No existing method simultaneously resolves the singularity problem, overcomes the small sample size restriction, and provides a closed-form, hardware-agnostic solution suitable for IoT gateways.

CPAI directly addresses this gap. By reformulating the discriminant analysis objective as an explicit target assignment problem, solving $(K + \lambda I)\alpha = \beta$ to force each class to a predefined, distinct projection coordinate, CPAI structurally prevents singularity collapse. Kernelization extends this to arbitrary data dimensionalities, removing the $n \leq d$ constraint. The closed-form solution requires no iterative optimization and no GPU, making CPAI uniquely aligned with the efficiency requirements of resource-constrained IoT security.

3. Our Proposed Approach

In this section, we present our proposed algorithm CPAI, detailing its theoretical foundations and implementation. We first introduce the key concepts underlying the algorithm, followed by a detailed description of its implementation. We then explain how we integrate kernel methods to overcome the small sample size problem, enhancing the approach's suitability for real-world IoT network security applications.

3.1. Linear Case

Suppose the dataset consists of n samples $\{x^i\}_{i=1}^n$ belonging to c classes, denoted as $\{x_1^i\}_{i=1}^{n_1}, \dots, \{x_c^i\}_{i=1}^{n_c}$, where n_j is the number of samples in the j^{th} class, and $x_j^i \in \mathbb{R}^d$ represents the i^{th} sample vector of the j^{th} class. The total number of samples is $n = \sum_{j=1}^c n_j$. Throughout this paper, we use the standard inner product $(x, y) = x^T y$, $\forall x, y \in \mathbb{R}^d$. The Euclidean norm or the length of vector x is denoted by $\|x\| = (x, x)^{\frac{1}{2}}$. The projection of the vector u onto a non-zero vector v is a scalar $(u, v)/\|v\|$.

This section primarily addresses the small-sample-size (SSS) case in the classical sense, where the number of data instances is less than or equal to the feature dimension (i.e., $n \leq d$), and assumes linear independence among sample vectors. The latter assumption is not merely a mathematical convenience but a fundamental prerequisite for the existence of a valid null space in NFST [15] and its subsequent extensions [19, 7, 3]. Consequently, our proposed method is established upon the following theorem.

Theorem 1. Suppose $\{u_1, \dots, u_n\} \subset \mathbb{R}^d$ are linearly independent. For any target projection vector $\beta = (b_1, \dots, b_n)^T \in$

$\mathbb{R}^n$, there exists a unique vector $\theta \in \text{span}\{u_1, \dots, u_n\} := \{\gamma_1 u_1 + \gamma_2 u_2 + \dots + \gamma_n u_n, \gamma_i \in \mathbb{R}, i = 1, n\}$ such that $(\theta, u_i) = b_i$ for all $i = 1, \dots, n$.

PROOF. Let $\beta = (b_1, \dots, b_n)^T \in \mathbb{R}^n$ be an arbitrary fixed target vector. The proof ends if the linear equation below admits a unique solution θ within the subspace $\text{span}\{u_1, \dots, u_n\}$:

$$Q^T \theta = \beta, \quad (1)$$

where $Q = (u_1, u_2, \dots, u_n) \in \mathbb{R}^{d \times n}$. Since we seek a solution $\theta \in \text{span}\{u_1, \dots, u_n\}$, it can be expressed as a unique linear combination of the basis vectors $\{u_1, \dots, u_n\}$ (as they are linearly independent)

$$\theta = \alpha_1 u_1 + \dots + \alpha_n u_n = Q\alpha, \quad (2)$$

where $\alpha = (\alpha_1, \dots, \alpha_n)^T \in \mathbb{R}^n$ is the unique vector of coefficients. Substituting (1) with its basis expansion (2), one obtains

$$Q^T Q \alpha = \beta. \quad (3)$$

The existence and uniqueness of the solution θ depend on the existence and uniqueness of the solution α to (3). It remains to prove that the $n \times n$ matrix $Q^T Q$ is of full rank (i.e., invertible). To this end, consider the homogeneous system $Q^T Q \alpha' = 0$. Multiplying by $(\alpha')^T$ on the left, we get $(\alpha')^T Q^T Q \alpha' = 0$, which implies $(Q\alpha', Q\alpha') = \|Q\alpha'\|^2 = 0$. This holds if and only if $Q\alpha' = 0$. Since the columns of Q (the vectors $u_1, \dots, u_n$) are linearly independent by assumption, the only solution to $Q\alpha' = 0$ is the trivial solution $\alpha' = 0$. It follows from linear algebra theory that $Q^T Q$ is of full rank and thus invertible.

Consequently, the system (3) has a unique solution α . Substituting this unique α back into (2) yields the unique solution θ within the subspace $\text{span}\{u_1, \dots, u_n\}$.

This theorem guarantees that, provided the linear independence of the sample vectors $\{x^i\}$, there exists a unique discriminant vector θ (within the span of these samples) that maps each training sample x_j^i to an arbitrarily predefined target value b_j . This allows us to force samples of the j^{th} class to map to a single target value (e.g., b_j), while samples from different classes (i.e., $k \neq j$) can be mapped to different target values ($b_k \neq b_j$). Notably, these target projection values can be chosen arbitrarily.

Lemma 2. Let H_j denote the affine hull generated by the data samples of the j^{th} class, i.e.,

$$H_j = \{x = a_1 x_j^1 + a_2 x_j^2 + \dots + a_{n_j} x_j^{n_j} : \sum_{i=1}^{n_j} a_i = 1\}.$$

Let θ be a vector constructed according to Theorem 1 such that $(\theta, x_j^i) = b_j$ (a constant value) for all samples $i = 1, \dots, n_j$ of the j^{th} class. Then, for any vector $x \in H_j$, it holds that $(\theta, x) = b_j$.

PROOF. The proof is straightforward. For any $x \in H_j$, one has the following linear combination: $x = a_1 x_j^1 + \dots + a_{n_j} x_j^{n_j}$ for some coefficients $a_1, \dots, a_{n_j}$ such that $\sum_{i=1}^{n_j} a_i = 1$. By the linearity of the inner product and the construction of θ , we have $(\theta, x_j^i) = b_j$ for all $i = 1, \dots, n_j$. Substituting this into the equation above yields:

$$(\theta, x) = \sum_{i=1}^{n_j} a_i b_j = b_j \sum_{i=1}^{n_j} a_i = b_j$$

provided that $(\theta, x_j^i) = b_j$ for any $i = 1, \dots, n_j$, and $\sum_{i=1}^{n_j} a_i = 1$.

The above lemma ensures that any point within the affine hull of the samples of the j^{th} class maps to the same predefined target value b_j when projected onto the discriminant direction θ .

According to Theorem (1), discriminant directions θ_i can be constructed by selecting appropriate target vectors β^i . A transformation matrix is then defined using one or more such vectors. Let $\Phi = (\theta_1, \dots, \theta_k)$ denote a matrix whose columns consist of k discriminant vectors. The transformed data is given by $y = \Phi^T x$. With appropriately chosen β^i vectors, this transformation maps samples of the same class to a single target point (or vector) in the k -dimensional projection space, while ensuring that target points for different classes are distinct. This framework leads to two primary implementation strategies for CPAI as a classifier:

1. **One-vs-All (OvA) approach:** This strategy classifies all c classes using a single discriminant vector θ . The target vector $\beta \in \mathbb{R}^n$ is constructed such that all samples belonging to the j^{th} class are mapped to the same target value b_j , while ensuring that target values for different classes remain distinct (i.e., $b_k \neq b_j$ for all $k \neq j$). For instance, β can be defined as:

$$\beta = (\underbrace{b_1, \dots, b_1}_{n_1}, \dots, \underbrace{b_j, \dots, b_j}_{n_j}, \dots, \underbrace{b_c, \dots, b_c}_{n_c})^T, \quad (4)$$

where $b_1, \dots, b_c$ are predefined, distinct scalar values that govern the final projection coordinates. This concept is illustrated in Fig. 1, where samples from multiple classes are projected onto a single discriminant direction θ , effectively mapping them to well-separated points $b_1, b_2, \dots, b_c$ within the 1D discriminant space.

2. **One-vs-Rest (OvR) approach:** This strategy decomposes the multi-class problem into multiple binary classification problems. A common variant is **One-vs-Rest (OvR)** where c separate discriminant vectors $\{\theta_1, \dots, \theta_c\}$ are constructed. For each j^{th} class, a discriminant vector θ_j is found such that samples of this class project to one value (e.g., 1), while samples from all other classes project to a different value (e.g., 0). This is achieved by setting c different target vectors $\beta^1, \dots, \beta^c$. For the j^{th} direction θ_j , the corresponding

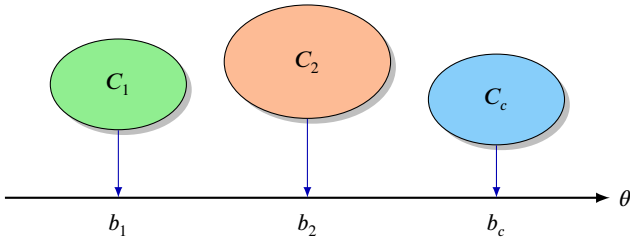


Figure 1: Visualization of the One-vs-All (OvA) strategy. Multiple class manifolds ($C_1, \dots, C_c$) are projected onto a single discriminant direction θ . By enforcing specific target coordinates b_j , the algorithm ensures that each class collapses into a distinct, non-overlapping point in the projected space.

target vector $\beta^j \in \mathbb{R}^n$ is predefined as:

$$(\beta^j)_i = \begin{cases} 1 & \text{if sample } i \in \text{class } j, \\ 0 & \text{if sample } i \notin \text{class } j. \end{cases}$$

Equivalently, the target vector is expressed as:

$$\beta^j = (\underbrace{0, \dots, 0}_{n_1}, \dots, \underbrace{1, \dots, 1}_{n_j}, \dots, \underbrace{0, \dots, 0}_{n_c})^T. \quad (5)$$

By explicitly enforcing these binary targets, we provide a structural safeguard against class merging (singularity), ensuring each class remains linearly separable in the discriminant space. This concept is visualized in Fig. 2, where multiple discriminant vectors are utilized to partition the feature space into distinct class-specific regions.

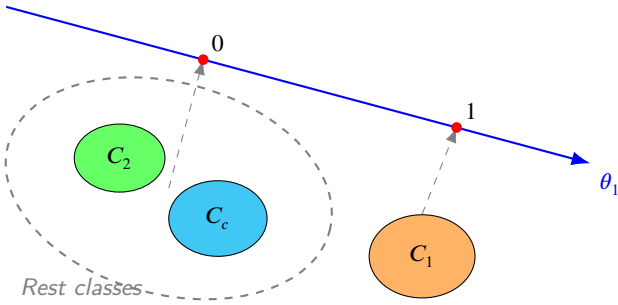


Figure 2: Illustration of the One-vs-Rest (OvR) strategy for class C_1 . The target class C_1 (orange) is projected onto the discriminant vector θ_1 to a specific target point 1, while the remaining classes ($C_2, \dots, C_c$, grouped by the dashed ellipse) are mapped to a different target 0. This explicit control of projection points prevents the class-merging singularity problem.

3.2. Nonlinear Case - Application of Kernel Trick

The linear CPAI formulation presented in Section 3.1 relies on the assumption of linear independence of the sample vectors x^i , typically necessitates $n \leq d$. Such assumptions may not hold in many real-world problems, particularly

when $n > d$ or the underlying class structures are non-linear, limiting the applicability of the linear theory. To overcome these limitations and to allow for more flexibility in the proposed model, the kernel trick can be employed, allowing the analysis to be performed implicitly in a potentially higher-dimensional Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$. We map features via a nonlinear function $\Psi : \mathcal{X} \rightarrow \mathcal{H}$, where the inner product is defined by a kernel function: $(\Psi(x^i), \Psi(x^l))_{\mathcal{H}} = \kappa(x^i, x^l)$.

To provide a rigorous justification for our kernelized formulation, we invoke the Representer Theorem [26]. This theorem states that for any regularized empirical risk functional defined over RKHS, the optimal solution can be expressed as a linear combination of the kernel evaluations at the training points. In the context of CPAI, the objective is to find a discriminant vector $\theta \in \mathcal{H}$ that satisfies $(\Psi(x^i), \theta)_{\mathcal{H}} = \beta_i$. According to the Representer Theorem, the optimal θ must lie within the span of the mapped training samples:

$$\theta = \sum_{i=1}^n \alpha_i \Psi(x^i). \quad (6)$$

This theoretical foundation allows us to transform the problem from an infinite-dimensional optimization in $\mathcal{H}$ into a finite-dimensional linear system:

$$K\alpha = \beta, \quad (7)$$

where K is the kernel matrix defined by:

$$K = [\kappa(x^i, x^l)]_{i,l \in \{1, \dots, n\}}. \quad (8)$$

The existence and uniqueness of the solution of Eq. (7) depend on the nonsingularity of the kernel matrix K . We recall that a kernel function κ is said to be strictly positive definite if its corresponding kernel matrix K is positive definite for any set of distinct data points. In such a case, the system is guaranteed to admit a unique solution. More precisely, a sufficient condition for a symmetric matrix K to be strictly positive definite is that it satisfies the strictly diagonally dominant (SDD) property, defined as $|K_{ii}| > \sum_{l \neq i} |K_{il}|$ for all i [14]. Consider the RBF kernel, $\kappa(x, y) = \exp(-\|x - y\|^2 / (2\sigma^2))$, with a bandwidth parameter $\sigma > 0$. By construction, K is a symmetric matrix with unit diagonal elements ($K_{ii} = 1$) and positive off-diagonal entries ($0 < K_{il} < 1$ for distinct samples $x_i \neq x_l$).

As σ decreases (or the bandwidth $\gamma = 1/2\sigma^2$ increases), the off-diagonal terms $\kappa(x_i, x_l)$ exponentially decay toward zero. Consequently, for a sufficiently small σ , the row sum $\sum_{l \neq i} K_{il}$ becomes strictly less than 1, rendering the matrix K strictly diagonally dominant and thus strictly positive definite. Similarly, for the Laplacian kernel $\kappa(x, y) = \exp(-\alpha\|x - y\|)$, setting the parameter α large enough ensures that the SDD property is maintained.

While the theoretical positive definiteness of the kernel matrix K ensures invertibility, it can still become numerically ill-conditioned in practice (i.e., having eigenvalues approaching zero). This leads to numerical instability and

potential class-merging singularities when solving the linear system. For instance, with the RBF kernel $\kappa(x, z) = \exp(-\gamma\|x - z\|^2)$, a very small bandwidth parameter γ causes the off-diagonal elements of K to approach 1, rendering the matrix nearly singular and sensitive to noise.

To address this instability and provide a structural safeguard against the singularity problem observed in NFST, a Tikhonov regularization term is introduced. This transforms the problem into a robust, regularized system, which is the final formulation solved in our CPAI implementation:

$$(K + \lambda I)\alpha = \beta, \quad (9)$$

where $\lambda > 0$ is a small regularization hyperparameter and I is the identity matrix. From a spectral perspective, the addition of the diagonal term λI shifts all eigenvalues of K by exactly λ . Since the eigenvalues of a symmetric positive semi-definite kernel matrix are non-negative ($\lambda_j \geq 0$), the eigenvalues of $(K + \lambda I)$ are strictly positive ($\lambda_j + \lambda > 0$). This ensures that the system is always strictly positive definite and well-conditioned, guaranteeing a unique, stable solution for α even in the presence of redundant or non-distinct IoT data samples.

3.3. CPAI vs. Kernel Ridge Regression

The regularized system in Eq. 9, while algebraically identical to Kernel Ridge Regression (KRR), is driven by fundamentally distinct motivations. KRR minimises a penalised prediction error with respect to given numerical labels, treating the target values as fixed regression outputs. CPAI, by contrast, treats those same target values as geometric coordinates that are deliberately chosen to enforce class separation in the projection space. Rather than merely distorting the Hilbert space $\mathcal{H}$, CPAI serves to uncover and linearize the latent, complex nonlinear dependencies within the features through the kernel trick. Within this framework, the Reproducing Kernel Hilbert Space (RKHS) is modeled as a topological landscape where legitimate traffic is mapped to a consistent altitude. This allows the model to identify anomalies not by mere numerical error, but as structural deviations that fail to reach the established plateau of normal class consensus.

3.4. Heuristic for Automatic Gamma Selection

The performance of kernel-based methods is critically dependent on the RBF kernel's bandwidth hyperparameter, γ . To avoid computationally prohibitive grid search, we propose an efficient data-driven heuristic. Our method estimates an optimal γ by relating it to the data's local structure via the average distance to the nearest neighbors. We compute the median of these average distances, σ , which serves as a robust estimator of the local data scale. The gamma is then computed as $\gamma = 1/(2\sigma^2)$. This heuristic calibrates the kernel's bandwidth to the intrinsic data geometry, enhancing the model's practicality.

3.5. Implementation of Kernel CPAI Algorithms

Based on the theoretical framework presented in Section 3.2, we detail the procedures for obtaining the discriminant

coefficients for both OvA and One-vs-Rest (OvR) strategies. Unlike the linear case, the kernelized CPAI directly computes the coefficient vector α , which defines the discriminant direction implicitly in the RKHS.

Algorithm 1: Kernel CPAI: One-vs-All (OvA)

Input: Q : Training data matrix; κ : Kernel function; γ, λ : Hyperparameters.

Output: α : Optimal coefficient vector.

- 1: **Construct** the kernel matrix $K \in \mathbb{R}^{n \times n}$ where $K_{il} = \kappa(x^i, x^l)$ using bandwidth γ .
- 2: **Initialize** the multi-class target vector:

$$\beta = (\underbrace{b_1, \dots, b_1}_{n_1}, \dots, \underbrace{b_j, \dots, b_j}_{n_j}, \dots, \underbrace{b_c, \dots, b_c}_{n_c})^T$$

- 3: Solve $(K + \lambda I)\alpha = \beta$ for α .

4: **return** α .

Algorithm 2 outlines the One-vs-Rest (OvR) strategy. This approach constructs c separate classifiers to isolate each class from the remaining data manifold.

Algorithm 2: Kernel CPAI: One-vs-Rest (OvR)

Input: Q : Training data matrix; κ, γ, λ .

Output: $\mathcal{A} = \{\alpha^1, \alpha^2, \dots, \alpha^c\}$: Set of coefficient vectors.

- 1: **Construct** the kernel matrix $K \in \mathbb{R}^{n \times n}$ using parameters γ .
- 2: **for** $j = 1$ to c **do**
- 3: **Construct** the binary target vector for class j :

$$\beta^j = (\underbrace{0, \dots, 0}_{n_1}, \dots, \underbrace{1, \dots, 1}_{n_j}, \dots, \underbrace{0, \dots, 0}_{n_c})^T$$

- 4: Solve the stable system $(K + \lambda I)\alpha^j = \beta^j$ for α^j .
 - 5: **end for**
 - 6: **return** $\mathcal{A}$.
-

Decision Rule: At inference time, a test sample x is classified by projecting it into the discriminant space via the decision function $f(x) = \sum_{i=1}^n \alpha_i \kappa(x, x^i)$. Depending on the training strategy, the final decision is made as follows:

- **Multiclass classification:** In the OvR framework, the sample is assigned to the class j whose discriminant function $f_j(x)$ yields the highest score. The predicted label $\hat{y}$ is determined by the argmax rule:

$$\hat{y} = \arg \max_{j \in \{1, \dots, c\}} f_j(x). \quad (10)$$

- **Anomaly identification:** CPAI provides a natural mechanism for identifying anomalies. Since normal samples are forced to collapse toward a specific target

b_{normal} , any test instance x whose projected value $f_{normal}(x)$ deviates significantly from this target-specificity, if $f_{normal}(x) < \tau$, where τ is a confidence threshold-is flagged as an anomaly. This structural deviation indicates that the sample does not conform to the learned manifold of legitimate network behavior.

4. Further Theoretical Analysis

To distinguish CPAI from generic regression frameworks and clarify its structural advantages over traditional null-space methods, we provide a deeper analysis of its geometric stability and spectral properties. For the sake of theoretical clarity, the analysis in this section focuses on the behavior of a single-class manifold, where the target vector is set as the all-ones vector ($\beta = 1$). In this context, any deviation of a new sample from this target indicates an anomaly.

4.1. Sample-size Invariant Stability

Let $y = [f(x^1), \dots, f(x^n)]^T = K\alpha$ be the outputs vector for the training samples. It follows from the CPAI solution that $y = 1 - \lambda\alpha$. The variance of the projected points is derived as:

$$\text{Var}(y) = \frac{\lambda^2}{n} \|\alpha - \bar{\alpha}1\|^2 \quad (11)$$

where $\bar{\alpha} = \frac{1}{n}1^T\alpha$. Furthermore, since $(K + \lambda I)$ is positive definite, the norm $\|\alpha\|$ is bounded by $\sqrt{n}/\lambda$. Consequently, $\text{Var}(y) \leq 1$. This bound holds regardless of the sample size (n), dimensionality (d), or the kernel eigenstructure.

4.2. Spectral Dynamics

Let the kernel matrix admit the eigendecomposition $K = \sum_{j=1}^n \lambda_j v_j v_j^T$, with $\{v_j\}$ forming an orthonormal basis. The CPAI solution admits the spectral expansion:

$$\alpha = \sum_{j=1}^n \frac{(1, v_j)}{\lambda_j + \lambda} v_j. \quad (12)$$

PROOF. To derive the spectral expansion of the CPAI coefficient vector α , we consider the regularized system defined in Eq. (9) with the unit target $\beta = 1$:

$$(K + \lambda I)\alpha = 1. \quad (13)$$

Based on the eigendecomposition of the kernel matrix $K = \sum_{j=1}^n \lambda_j v_j v_j^T$ and the identity $I = \sum_{j=1}^n v_j v_j^T$, the operator $(K + \lambda I)$ can be expressed in its spectral form as:

$$K + \lambda I = \sum_{j=1}^n (\lambda_j + \lambda) v_j v_j^T. \quad (14)$$

Since K is a symmetric positive semi-definite matrix ($\lambda_j \geq 0$) and $\lambda > 0$, the inverse of this operator is well-defined and

computed by inverting the shifted eigenvalues:

$$(K + \lambda I)^{-1} = \sum_{j=1}^n \frac{1}{\lambda_j + \lambda} v_j v_j^T. \quad (15)$$

Applying this inverse to the target vector 1 yields:

$$\begin{aligned} \alpha &= (K + \lambda I)^{-1} 1 \\ &= \left(\sum_{j=1}^n \frac{1}{\lambda_j + \lambda} v_j v_j^T \right) 1 \\ &= \sum_{j=1}^n \frac{1}{\lambda_j + \lambda} v_j (v_j^T 1) \\ &= \sum_{j=1}^n \frac{(1, v_j)}{\lambda_j + \lambda} v_j. \end{aligned}$$

Notice that the term $S_j := (1, v_j)$ represents the projection of the data centroid onto the principal axes of the manifold in the RKHS.

4.2.1. Centroid-to-Axis Alignment Theorem

Recall $\mathcal{H}$ as the RKHS induced by the kernel $\kappa(\cdot, \cdot)$ and Ψ as the implicit feature map. The kernel mean element of the class is $\mu = \frac{1}{n} \sum_{k=1}^n \Psi(x^k)$. Analogous to Kernel Principal Component Analysis (KPCA) [28], the j^{th} principal axis $u_j \in \mathcal{H}$ is constructed as $u_j = \frac{1}{\sqrt{\lambda_j}} \sum_{i=1}^n v_{j,i} \Psi(x^i)$.

Theorem 3. *The projection of the centroid μ onto the j^{th} principal axis u_j in the RKHS is proportional to S_j :*

$$(\mu, u_j)_{\mathcal{H}} = \frac{\sqrt{\lambda_j}}{n} S_j. \quad (16)$$

PROOF. By substituting the definitions of μ and u_j :

$$\begin{aligned} (\mu, u_j)_{\mathcal{H}} &= \left(\frac{1}{n} \sum_{k=1}^n \Psi(x^k), \frac{1}{\sqrt{\lambda_j}} \sum_{i=1}^n v_{j,i} \Psi(x^i) \right)_{\mathcal{H}} \\ &= \frac{1}{n\sqrt{\lambda_j}} \sum_{k=1}^n (K v_j)_k. \end{aligned}$$

Since v_j is an eigenvector of K ($K v_j = \lambda_j v_j$), we have:

$$(\mu, u_j)_{\mathcal{H}} = \frac{\lambda_j}{n\sqrt{\lambda_j}} \sum_{k=1}^n v_{j,k} = \frac{\sqrt{\lambda_j}}{n} S_j. \quad (17)$$

4.2.2. Structural Implication

To clarify the physical orientation of the learned model, we examine the discriminant vector θ in the RKHS. By combining the spectral expansion with the construction of principal axes u_j , we derive the following:

$$\theta = \sum_{j=1}^n \left(\frac{\sqrt{\lambda_j} S_j}{\lambda_j + \lambda} \right) u_j. \quad (18)$$

This derivation provides a profound justification for the CPAI solution. It acts as a structural filter:

- $S_j \neq 0$: These axes are well-aligned with the data centroid μ , representing global invariant features. CPAI preserves these channels.
- $S_j \approx 0$: These axes represent local noise or sample-specific variations orthogonal to the centroid. CPAI effectively nullifies these directions.

4.3. Local Manifold Preservation

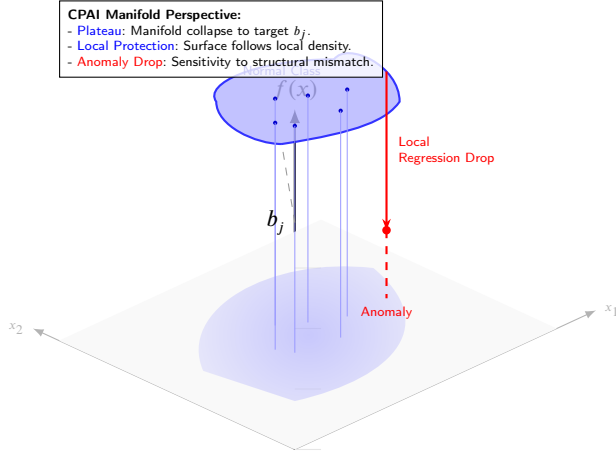


Figure 3: Geometric mechanism of CPAI: Normal samples are lifted to a target plateau at b_j , while a local anomaly suffers an altitude drop due to structural mismatch.

As illustrated in Fig. 3, CPAI transforms the input feature space into a topological landscape where normal data clusters form a *Plateau* at height b_j . A key theoretical property of this mapping is that $f(x)$ is globally bounded and strictly below the target b_j , which we prove as follows:

Theorem 4. *For any $\lambda > 0$ and a positive definite kernel κ , the CPAI prediction function $f(x)$ for a single class with target $b_j = 1$ satisfies $f(x) < 1$ for all $x \in \mathbb{R}^d$.*

PROOF. Consider the CPAI system $(K + \lambda I)\alpha = 1$. Let $K = V\Lambda V^T$ be the eigendecomposition of the kernel matrix, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i \geq 0$. The prediction at any point x is given by $f(x) = \mathbf{k}(x)^T \alpha$, where $\mathbf{k}(x) = [\kappa(x, x_1), \dots, \kappa(x, x_n)]^T$. Substituting the solution for α :

$$f(x) = \mathbf{k}(x)^T V(\Lambda + \lambda I)^{-1} V^T \mathbf{1} = \sum_{j=1}^n \frac{(\mathbf{k}(x)^T v_j)(v_j^T \mathbf{1})}{\lambda_j + \lambda} \quad (19)$$

At a training point x_i , $\mathbf{k}(x_i)$ is the i -th column of K , thus $\mathbf{k}(x_i)^T v_j = \lambda_j v_{j,i}$. The prediction becomes:

$$f(x_i) = \sum_{j=1}^n \left(\frac{\lambda_j}{\lambda_j + \lambda} \right) v_{j,i} (v_j^T \mathbf{1}) \quad (20)$$

Since $\lambda > 0$, the scaling factor satisfies $\frac{\lambda_j}{\lambda_j + \lambda} < 1$ for all j . Furthermore, since V is an orthogonal matrix,

$\sum_{j=1}^n v_{j,i} (v_j^T \mathbf{1}) = [V V^T \mathbf{1}]_i = [\mathbf{1}]_i = 1$. It follows that:

$$f(x_i) < \sum_{j=1}^n 1 \cdot v_{j,i} (v_j^T \mathbf{1}) = 1 \quad (21)$$

Because the kernel $\kappa(x, y)$ decays as x moves away from the training density, the maximum value of $f(x)$ is anchored at the training manifold. Therefore, $f(x) < 1$ holds globally.

5. Experiments

This section outlines our experiments designed to address the following research questions:

- **RQ1:** How effectively does the CPAI model classify attacks under two training approaches?
- **RQ2:** How significantly do the training approach and kernel selection impact the classification performance?
- **RQ3:** How does the performance of the CPAI model compare to baseline models?

5.1. Datasets

To assess the efficacy of our proposed algorithm, we conduct a comprehensive evaluation across five widely-used datasets: BoT-IoT, IoTID20, ToN-IoT, N-BaIoT, and CIC-IoT2023. These datasets are specifically designed to emulate the complexities of IoT network environments and encompass various attack traffic patterns. A concise overview of the utilised datasets is provided below:

BoT-IoT [18]: The BoT-IoT dataset, published by the Cyber Range Lab of UNSW Canberra, serves as an important resource for supporting research in IoT network security. This dataset comprises various network traffic data captured from IoT devices infected with diverse botnet malware strains. These malicious programs are known to initiate a wide spectrum of cyberattacks, encompassing distributed denial of service (DDoS), brute-force incursions, and reconnaissance scans. The network traffic data acquisition took place within a meticulously controlled environment that replicates real-world IoT network configurations. Through comprehensive analysis, a comprehensive dataset of over 20 features was extracted from the raw network traffic, offering insights into key network parameters, flow dynamics, and traffic volume.

IoTID20 [30]: The IoTID20 dataset is well-designed to facilitate the development of IDSs for the IoT environment. It simulates real-world network traffic in an IoT setup and provides a foundation for identifying and mitigating malicious activities. IoTID20 covers a wide range of cyberattacks that specifically target IoT devices and networks. These include Distributed Denial of Service (DDoS), man-in-the-middle (MITM), data exfiltration, and scanning attacks.

ToN-IoT [4]: The ToN-IoT dataset, provided by the Cyber Range Lab of UNSW Canberra, is a resource designed to facilitate the development of artificial intelligence-driven security solutions tailored for industrial IoT networks.

Unlike the BoT-IoT dataset, which focuses on attack traffic, the ToN-IoT dataset primarily encompasses benign network traffic generated by legitimate IoT devices operating under normal, non-compromised conditions. Furthermore, the ToN-IoT dataset also provides diversity, encompassing a wide range of IoT applications and communication protocols commonly deployed in real-world IIoT environments.

N-BaIoT [20]: The N-BaIoT dataset is a valuable resource for analysing the normal behavioural patterns of different IoT devices, encompassing sensors, actuators, and smart home appliances. By capturing the network traffic generated by these devices under benign conditions, the dataset establishes a baseline for comparison, aiding in identifying malicious activity. Moreover, the N-BaIoT dataset is notable for covering a wide range of IoT applications and communication protocols. This diversity ensures that the dataset is representative of real-world IoT deployments, enhancing its relevance and applicability to security research. The N-BaIoT dataset also includes instances of botnet attacks originating from two prominent botnet families: Mirai and Bashlite. These attacks provide valuable insights into the tactics and techniques employed by malicious actors seeking to exploit vulnerabilities in IoT devices.

CIC-IoT2023 [22]: The CIC-IoT2023 dataset, published by the Canadian Institute for Cybersecurity (CIC), represents a comprehensive resource designed to evaluate the performance and effectiveness of IDSs specifically tailored for IoT environments. This dataset includes a diverse range of IoT devices and simulates many cyberattacks that pose significant threats to IoT ecosystems. For the aim of this research, the attacks contained within the CIC-IoT2023 dataset have been systematically categorized into five distinct attack categories: DDoS/DoS, scan, web attacks, spoofing, and botnet attacks.

5.2. Evaluation metrics

To evaluate the detection performance of our proposed algorithm in classification cyber attacks, we use a comprehensive set of benchmarking metrics calculated based on the confusion matrix. These include Accuracy (ACC), True Positive Rate (TPR), Positive Predictive Value (PPV), and False Positive Rate (FPR), which are commonly used to evaluate model performance. While these metrics provide an overall assessment of correctness, their reliability can be compromised in imbalanced datasets. To address this limitation, the F1-score, which balances precision and recall, and the Matthews Correlation Coefficient (MCC), which considers all four confusion matrix components, are incorporated to provide a more robust evaluation. We describe the formula for each metric below:

- Accuracy (ACC):

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

- Recall - True Positive Rate (TPR):

$$TPR = \frac{TP}{TP + FN}$$

- False Alarm Rate - False Positive Rate (FPR):

$$FPR = \frac{FP}{FP + TN}$$

- Precision - Positive Predictive Value (PPV):

$$PPV = \frac{TP}{TP + FP}$$

- F1-score:

$$F1\text{-score} = \frac{2 * TPR * PPV}{TPR + PPV}$$

- Matthews Correlation Coefficient (MCC):

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

5.3. Baseline competitive models

In this research, we employ a comprehensive set of baseline algorithms to evaluate our proposed approach. These baselines are machine learning algorithms based on discriminant analysis approaches, including Gaussian Naive Bayes (GauNB), Kernel Null Foley-Sammon Transformation (KNFST), K-Nearest-Neighbour (KNN), Linear Discriminant Analysis (LDA), Nearest Centroid (NC), Support Vector Machine (SVM), Radius Neighbors Classifier (RNC) or Stochastic Gradient Descent-SVM (SGD). These algorithms were specifically selected for their proven robustness in handling high-dimensional data and their widespread use in network intrusion detection. For implementation, we utilize the Sklearn Python library for most models, with the exception of KNFST, for which we provide our own implementation based on the original algorithm [7], due to the unavailability of official source code. Through this diverse set of baseline models, we aim to thoroughly evaluate our approach against established methods and highlight our effectiveness in addressing the research problem.

5.4. Multiclass classification performance under two training approaches with different kernel

This subsection presents the comprehensive experimental results to solve **RQ1** and **RQ2**. Our analysis is structured in two parts: First, Tables 2 and 3, respectively, summarize the comparative performance of various kernels with two proposed approaches. These results indicate the high accuracy of CPAI in classifying attacks in IoT networks. Second, Figures 4 and 5 provide a deeper comparative analysis, specifically highlighting the mean MCC and mean F1-score across the different kernels between the two proposed training approaches. The results reveal the significant impact on choosing training approach and selecting kernel. The detailed results are presented as follows.

5.4.1. The classification performance using different kernels

Table 2 highlights the remarkable accuracy achieved by the CPAI when trained with the One-vs-All (OvA) strategy.

Table 2

Performance evaluation of CPAI's One-vs-All (OvA) approach across evaluation datasets, comparing different kernel settings using accuracy (ACC), true positive rate (TPR), and false positive rate (FPR) metrics. The best score is highlighted with red colour while the second best is highlighted with blue colour.

Kernel	BoT-IoT			IoTID20			ToN-IoT			N-BaIoT			CIC-IoT2023		
	ACC	TPR	FPR	ACC	TPR	FPR	ACC	TPR	FPR	ACC	TPR	FPR	ACC	TPR	FPR
None	98.39	94.48	0.97	82.86	62.00	10.71	88.18	51.78	6.76	96.93	86.17	1.73	91.12	57.94	5.33
Abel	99.24	97.94	0.45	95.24	88.47	2.98	98.02	80.50	1.11	98.97	85.53	0.57	95.21	69.70	2.79
Laplacian	99.49	84.26	0.30	97.57	77.78	1.46	98.23	81.44	1.00	99.42	87.54	0.32	95.41	69.41	2.68
Linear	98.32	94.35	1.01	83.64	60.42	10.23	89.48	56.31	6.01	95.58	80.11	2.49	91.63	61.33	5.02
Poly	99.23	97.25	0.46	88.18	71.98	7.39	95.92	84.21	2.33	98.94	95.22	0.60	93.28	69.99	4.03
Rbf	99.37	97.91	0.38	90.12	76.88	6.18	96.73	86.47	1.87	99.40	97.28	0.34	93.53	70.67	3.88
Sigmoid	92.36	78.70	4.58	78.20	43.51	13.63	87.20	45.75	7.31	90.57	57.56	5.31	85.68	39.44	8.59
Sobolev	95.03	88.73	2.98	78.92	52.09	13.18	89.67	62.88	5.90	92.70	63.96	4.10	91.67	50.29	5.21

The experiment results reveal that three positive definite kernels, including Abel, Laplacian, and RBF, alternately achieve the top accuracy, but none of the kernels dominates the others. Particularly, the CPAI-OvA approach consistently achieves the highest accuracy from 95.41% across all benchmark datasets. Peak performance is observed on the N-BaIoT and BoT-IoT datasets, with accuracies of 99.42% and 99.49%, respectively, using the kernel Laplacian. The lowest accuracy, though still high, is recorded in the CIC-IoT2023 dataset, at 95.41%. While the Laplacian kernel achieves the highest accuracy in all datasets, it is surpassed by the kernel Abel and RBF in terms of TPR. The inequivalence between TPR and ACC suggests that Laplacian may be sensitive to class imbalance and biased by the majority class. Additionally, the Abel and Rbf kernels also alternately archive the best TPR score across benchmarked datasets. In contrast, non-positive definite kernels such as Sigmoid and Linear exhibit significantly lower ACC and TPR, along with higher False Positive Rates (FPR), compared to most positive definite kernels.

Table 3 highlights the superior accuracy achieved by the CPAI when trained with the **One-vs-Rest (OvR)** strategy. The experiment results indicate that the outstanding Laplacian kernel surpasses other kernels in all benchmarked datasets. Notably, the **CPAI-OvR** approach consistently surpasses 98.42% accuracy across all benchmark datasets when employing the Laplacian kernel. The peak performance is observed on the N-BaIoT and BoT-IoT datasets, achieving accuracies of 99.96% and 99.77%, respectively. The lowest accuracy is recorded in the IoTID20 dataset, at 98.42%, which is close to the highest accuracy. Furthermore, the **CPAI-OvR** -Laplacian approach demonstrates stability, consistently achieving the best TPR while maintaining the lowest FPR across all datasets. Beyond the impressive accuracy of the Laplacian kernel, the Abel and RBF kernels alternately achieve the second-best accuracy and TPR on most benchmarked datasets. Additionally, the experiment results reveal that non-positive definite kernels, such as Sigmoid and Linear, exhibit significantly lower ACC and TPR, along with higher FPR, compared to most positive definite kernels.

5.4.2. The classification performance comparison between two training approach

Figures 4 and 5 show that the **CPAI-OvR** approach consistently outperforms the CPAI-OvA approach across all kernels in terms of MCC and F1-score. Specifically, while both approaches achieve the best MCC score using the Laplacian kernel, the **CPAI-OvR** significantly surpasses the OvA approach, with a score of 0.97 compared to 0.91, respectively. The largest gap was presented by the Sobolev kernel, with a difference of approximately 0.34, while the smallest gap was presented by the Abel kernel, with a difference of 0.05. This trend of OvR superiority is reflected in terms of F1-score, where **CPAI-OvR** surpasses CPAI-OvA by the largest gap of approximately 37% and the smallest gap of 9.8%. This result indicates the superiority of the OvR approach over the OvA approach in terms of classification performance.

Besides selecting training approaches, Figures 4 and 5 also indicate the significant impact of choosing the kernel. In general, selecting kernel has a significant effect on accuracy and consistently affects both training strategies similarly. The positive-define kernels, such as Abel, Laplacian, Rbf give higher accuracy than non-positive-define kernels like Linear and Sigmoid kernels. While the Laplacian archives the best MCC with 0.97 when trained with the OvR approach, the Sigmoid kernel only reaches an MCC of 0.72. This trend is similar to the OvA approach. Besides, applying positive-define kernels also gives an improvement in accuracy compared to not using kernels. However, the results also reveal that the non-positive-define kernels tend to give worse classification performance than not use kernels. These findings collectively highlight the crucial role of kernel selection in enhancing model performance.

5.5. Baseline comparison results

This subsection presents a comprehensive comparative analysis of our proposed algorithm against widely used baseline models on different evaluation metrics, directly answering the **RQ3**. The 4 presents the comparison results on dataset Bot-IoT, IoTID20, ToN-IoT, N-BaIoT and CICIoT2023, respectively. To facilitate interpretation, we highlight the highest-performing algorithm in each metric with red text and the second-best performer with blue text. The analysis demonstrates that our proposed algorithm

Table 3

Performance evaluation of CPAI's **One-vs-Rest (OvR)** approach across evaluation datasets, comparing different kernel settings using accuracy (ACC), true positive rate (TPR), and false positive rate (FPR) metrics. The best score is highlighted with red colour while the second best is highlighted with blue colour.

Kernel	BoT-IoT			IoTID20			ToN-IoT			N-BaIoT			CIC-IoT2023		
	ACC	TPR	FPR	ACC	TPR	FPR	ACC	TPR	FPR	ACC	TPR	FPR	ACC	TPR	FPR
None	99.51	98.52	0.30	91.08	78.31	5.58	98.23	91.75	1.01	98.38	93.57	0.91	97.25	86.57	1.65
Abel	99.62	98.59	0.23	95.98	89.91	2.51	99.05	95.80	0.54	99.95	99.78	0.03	97.93	89.89	1.24
Laplacian	99.77	99.14	0.14	98.42	96.05	0.99	99.24	96.96	0.43	99.96	99.84	0.02	98.83	94.25	0.70
Linear	99.54	98.48	0.28	92.94	82.99	4.41	98.24	92.06	1.00	96.90	81.03	1.74	97.90	89.71	1.26
Poly	99.37	97.60	0.38	93.34	84.53	4.16	98.45	93.26	0.89	99.78	99.00	0.13	98.32	91.86	1.01
Rbf	99.62	98.59	0.23	94.64	86.52	3.35	99.06	95.97	0.54	99.86	99.40	0.08	98.27	91.57	1.04
Sigmoid	97.02	90.88	1.79	86.76	66.23	8.28	95.13	77.97	2.78	94.81	77.85	2.92	91.02	53.23	5.39
Sobolev	99.41	97.95	0.36	95.74	90.29	2.66	98.70	94.59	0.74	99.69	98.64	0.17	96.93	85.99	1.84

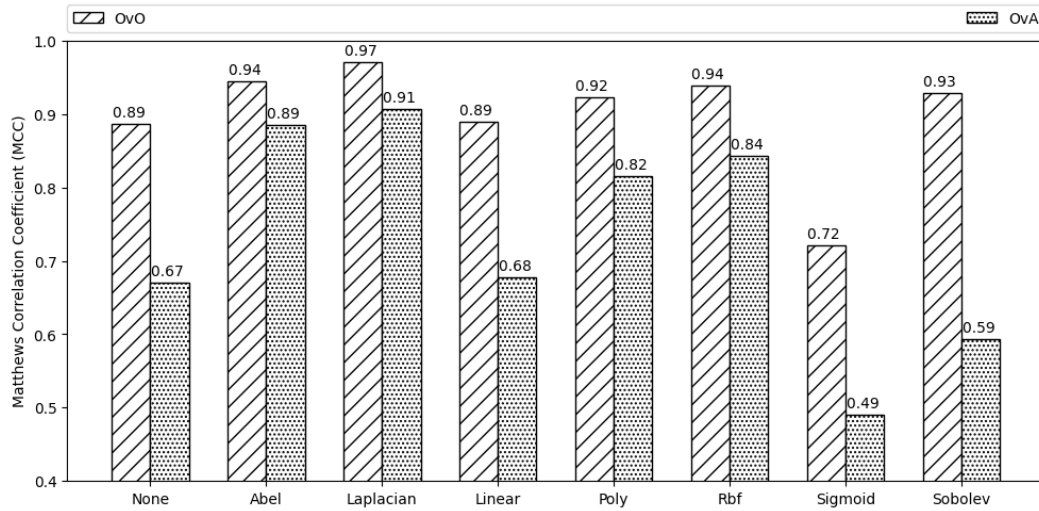


Figure 4: Comparative analysis of mean MCC score using different kernel functions between two CPAI's classification strategy approaches.

consistently achieves superior performance across multiple metrics compared to the baseline approaches.

For more details, training the proposed algorithm under the OvR strategy consistently yields the highest and most stable accuracy across all benchmark datasets. The

algorithm reaches its peak on the N-BaIoT dataset with MCC of 0.998, while maintaining a minimum MCC value exceeding 0.95. It also demonstrates stability by keeping the ACC above 98% in all datasets. Similarly, the **CPAI-OvR** approach achieves high scores across other metrics,

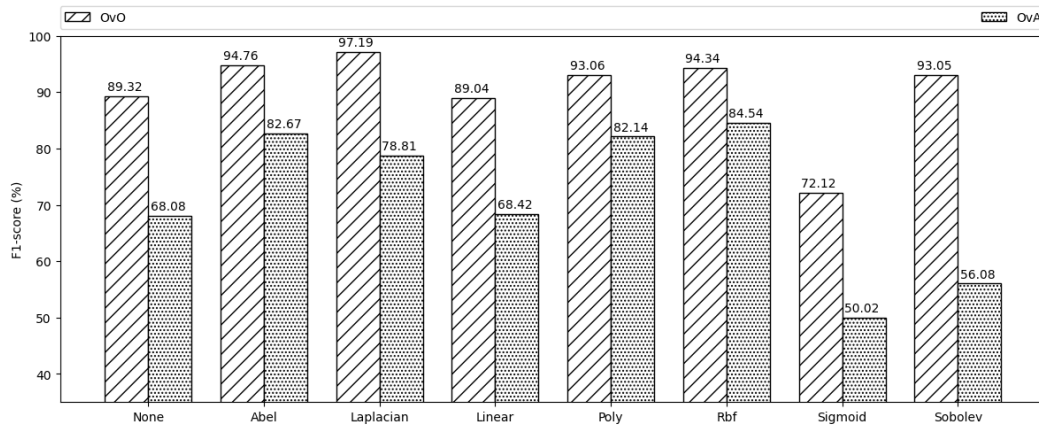


Figure 5: Comparative analysis of mean F1-score using different kernel functions between two CPAI's classification strategy approaches.

Table 4

Comparative evaluation of CPAI (OvA and OvR variants) against eight baseline models (GauNB, KNFST, KNN, LDA, NC, SVM, RNC, SGD) on the BoT-IoT dataset using six performance metrics: Matthews Correlation Coefficient (MCC), Accuracy (ACC), True Positive Rate (TPR), False Positive Rate (FPR), Positive Predictive Value (PPV), and F1-score.

Model	MCC ↑	ACC ↑	TPR ↑	FPR ↓	PPV ↑	F1 ↑
GauNB	0.796	93.99	81.59	3.61	83.16	78.71
KNFST	0.983	99.52	98.41	0.29	98.07	98.23
KNN	0.978	99.41	97.71	0.36	98.06	97.88
LDA	0.950	98.62	95.29	0.83	95.27	95.26
NC	0.703	91.63	64.73	5.02	72.74	61.35
SVM	0.916	97.68	93.02	1.39	92.61	92.68
RNC	0.925	97.89	92.83	1.26	95.59	93.78
SGD	0.976	99.34	97.75	0.40	97.65	97.69
CPAI-OvA	0.978	99.49	84.26	0.30	83.85	84.04
CPAI-OvR	0.992	99.77	99.14	0.14	99.19	99.17

Table 5

Comparative evaluation of CPAI (OvA and OvR variants) against eight baseline models (GauNB, KNFST, KNN, LDA, NC, SVM, RNC, SGD) on the IoTID20 dataset using six performance metrics: Matthews Correlation Coefficient (MCC), Accuracy (ACC), True Positive Rate (TPR), False Positive Rate (FPR), Positive Predictive Value (PPV), and F1-score.

Model	MCC ↑	ACC ↑	TPR ↑	FPR ↓	PPV ↑	F1 ↑
GauNB	0.505	81.06	52.65	11.84	77.73	49.06
KNFST	0.654	88.90	72.25	6.94	71.98	71.96
KNN	0.754	92.12	80.30	4.93	80.25	80.23
LDA	0.795	93.38	83.45	4.14	84.19	83.48
NC	0.432	81.50	53.75	11.56	52.45	49.41
SVM	0.788	93.20	83.00	4.25	83.62	83.13
RNC	0.419	78.46	46.15	13.46	67.09	43.57
SGD	0.653	87.28	68.20	7.95	80.57	67.08
CPAI-OvA	0.852	95.24	88.47	2.98	88.10	88.12
CPAI-OvR	0.951	98.42	96.05	0.99	96.05	96.04

surpassing 94% on all datasets, while keep the false positive rate remains consistently below 1%, lower than other competitors.

Beyond superior accuracy, CPAI-OvR flexibly adapts to different environments by reaching the first rank in all evaluated datasets. Specifically, the algorithm can accurately classify different attacks across all benchmark datasets, maintaining an MCC above 0.95. In contrast, the KNN algorithm achieved second place on only two datasets, while KNFST, and SVM sequence attained second positions on BoT-IoT, and CIC-IoT2023, respectively, but faltered on other datasets.

In conclusion, the comparison results highlight the outstanding of the proposed algorithm in classifying cyber attacks in the IoT networks. The CPAI-OvR consistently provides high accuracy while maintaining stability across different environments.

6. Conclusion and perspectives

This paper introduced CPAI, a novel discriminant analysis framework specifically engineered for anomaly detection in resource-constrained IoT environments. By explicitly controlling projection targets and leveraging the kernel trick, CPAI fundamentally resolves the long-standing singularity problem inherent in prior null-space methods. Our

theoretical analysis, complemented by spectral dynamics, establishes that CPAI provides a bounded and geometrically stable projection space, ensuring a high-confidence consensus for legitimate traffic manifolds. The extensive empirical evaluation across five large-scale IoT benchmarks demonstrates CPAI's superior performance, consistently achieving higher accuracy and stability compared to existing distance-based algorithms. The findings emphasized that CPAI bridges the gap between the efficiency of shallow methods and the separation power of supervised discriminant analysis, making it uniquely suited for resource-constrained IoT gateways.

However, our approach has several limitations. First, the performance of Kernel CPAI is dependent on the selection of a suitable kernel and its hyperparameters. While our proposed gamma heuristic mitigates this, the regularization parameter λ may still require tuning for optimal performance. Second, the kernelized version's training complexity scales with the number of samples (n), which can be a bottleneck for extremely large datasets.

The research findings open several promising avenues for future research and practical applications:

Table 6

Comparative evaluation of CPAI (OvA and OvR variants) against eight baseline models (GauNB, KNFST, KNN, LDA, NC, SVM, RNC, SGD) on the **ToN-IoT** dataset using six performance metrics: Matthews Correlation Coefficient (MCC), Accuracy (ACC), True Positive Rate (TPR), False Positive Rate (FPR), Positive Predictive Value (PPV), and F1-score.

Model	MCC ↑	ACC ↑	TPR ↑	FPR ↓	PPV ↑	F1 ↑
GauNB	0.277	83.53	38.21	9.41	33.55	27.32
KNFST	0.896	97.75	90.76	1.29	89.72	89.87
KNN	0.943	98.79	94.35	0.69	94.89	94.59
LDA	0.908	98.03	92.06	1.12	90.94	91.43
NC	0.259	82.66	39.43	9.91	40.41	25.33
SVM	0.875	97.32	87.33	1.53	89.27	88.00
RNC	0.871	97.15	86.58	1.63	93.78	88.91
SGD	0.829	96.30	83.57	2.11	85.50	83.64
CPAI-OvA	0.907	98.23	81.44	1.00	81.38	81.32
CPAI-OvR	0.965	99.24	96.96	0.43	96.61	96.76

Table 7

Comparative evaluation of CPAI (OvA and OvR variants) against eight baseline models (GauNB, KNFST, KNN, LDA, NC, SVM, RNC, SGD) on the **N-BalIoT** dataset using six performance metrics: Matthews Correlation Coefficient (MCC), Accuracy (ACC), True Positive Rate (TPR), False Positive Rate (FPR), Positive Predictive Value (PPV), and F1-score.

Model	MCC ↑	ACC ↑	TPR ↑	FPR ↓	PPV ↑	F1 ↑
GauNB	0.721	94.25	74.11	3.24	75.22	71.26
KNFST	0.960	99.20	96.39	0.45	96.69	96.36
KNN	0.984	99.68	98.56	0.18	98.57	98.55
LDA	0.883	97.62	89.28	1.34	90.39	87.82
NC	0.450	88.54	48.44	6.44	38.95	39.74
SVM	0.976	99.53	97.89	0.26	97.92	97.89
RNC	0.697	93.37	70.17	3.73	89.14	72.62
SGD	0.912	98.22	92.00	1.00	92.67	91.79
CPAI-OvA	0.969	99.40	97.28	0.34	97.32	97.27
CPAI-OvR	0.998	99.96	99.84	0.02	99.83	99.83

- Integrating scalable kernel approximation techniques, such as the Nyström method or Random Fourier Features, to enable real-time learning in massive-scale IoT data streams.
- In network security, understanding why a flow is flagged is as critical as detection accuracy. Subsequent studies could explore the inverse mapping of CPAI's discriminant directions to identify the specific network features contributing to the structural regression drop, providing administrators with actionable insights into attack vectors.
- IoT traffic is inherently dynamic. Developing an incremental or online version of CPAI that can update its structural plateau without retraining on the entire history would enhance its resilience against concept drift and evolving zero-day attacks.

Table 8

Comparative evaluation of CPAI (OvA and OvR variants) against eight baseline models (GauNB, KNFST, KNN, LDA, NC, SVM, RNC, SGD) on the **CIC-IoT2023** dataset using six performance metrics: Matthews Correlation Coefficient (MCC), Accuracy (ACC), True Positive Rate (TPR), False Positive Rate (FPR), Positive Predictive Value (PPV), and F1-score.

Model	MCC ↑	ACC ↑	TPR ↑	FPR ↓	PPV ↑	F1 ↑
GauNB	0.790	94.95	75.48	3.03	75.45	72.71
KNFST	0.836	96.17	81.17	2.30	80.97	81.00
KNN	0.858	96.67	83.47	2.00	83.88	83.48
LDA	0.773	94.68	73.55	3.19	74.29	73.46
NC	0.492	85.78	55.72	8.53	59.23	48.96
SVM	0.884	97.30	86.57	1.62	88.78	87.49
RNC	0.538	89.63	48.17	6.22	50.06	46.75
SGD	0.781	94.80	74.07	3.12	76.92	73.57
CPAI-OvA	0.771	95.41	69.41	2.68	63.07	63.82
CPAI-OvR	0.950	98.83	94.25	0.70	94.17	94.14

- Beyond intrusion detection, the principle of target-controlled manifold collapse can be extended to other one-class classification domains, such as industrial sensor fault diagnosis, medical image anomaly detection, and fraud detection in financial transactions.

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A. Overview of NFST and KNFST

In this appendix, we provide a brief overview of NFST and its kernelized version (KNFST) to make the paper self-contained and to provide context for the proposed CPAI method. Assume the dataset consists of n samples (denoted by x^i) belonging to c classes $\{x_1^i\}_{i=1}^{n_1}, \dots, \{x_c^i\}_{i=1}^{n_c}$ where n_j and $x_j^i \in \mathbb{R}^d$ denote the sample size and the i^{th} instance of the j^{th} class, respectively. Let us define $\bar{x}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} x_j^i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x^i$ as the mean of the j^{th} class and the total mean of the data, respectively. Foley-Sammon Transform (FST) [13] aims to find an optimal discriminant vector $\varphi \in \mathbb{R}^d$ that maximizes the Fisher criterion [12], under orthonormal

constraints:

$$J(\varphi) = \frac{\varphi^T S_b \varphi}{\varphi^T S_w \varphi}, \quad (22)$$

where $S_w \in \mathbb{R}^{d \times d}$ and $S_b \in \mathbb{R}^{d \times d}$ are the within-class scatter matrix and between-class scatter matrices, respectively, defined as

$$S_w := \sum_{j=1}^c \sum_{i=1}^{n_j} (x_j^i - \bar{x}_j)(x_j^i - \bar{x}_j)^T$$

$$S_b := \sum_{j=1}^c n_j (\bar{x}_j - \bar{x})(\bar{x}_j - \bar{x})^T.$$

If S_w is nonsingular, φ_1 is the unit eigenvector corresponding to the maximum eigenvalue of $S_w^{-1} S_b$. The remaining discriminant directions are obtained by solving

$$\max_{\varphi_i} J(\varphi) \quad \text{s.t., } \varphi_i^T \varphi_j = \delta_{ij}, \quad j = 1, \dots, i-1. \quad (23)$$

In contrast to FST, Null Foley-Sammon Transform (NFST) [15] aims to find φ —a so-called null projecting direction (NPD)—which satisfies:

$$\varphi^T S_w \varphi = 0, \quad (24)$$

$$\varphi^T S_b \varphi > 0. \quad (25)$$

Then the data have the best separability on the direction φ according to the Fisher discriminant criterion. For the **small-sample-size (SSS) case** (i.e., $n \leq d$), we additionally assume that the sample vectors are **linearly** independent. Let us define $Z_t = \{x \in \mathbb{R}^d \mid S_t x = 0\}$ where $S_t = S_w + S_b$ is the total scatter matrix, and $Z_t^\perp$ denotes the orthogonal complement space of Z_t . It can be shown that

Theorem 5. (see [15]) *If the sample vectors are linearly independent, there are $c-1$ NPDs $\varphi_i \in (Z_w \cap Z_t^\perp)$ where $Z_w = \{x \in \mathbb{R}^d \mid S_w x = 0\}$.*

To make it efficient for $d > n$ case, the nonlinear version of the theory was proposed; see the details in [19]. The key point is to take each column vector of K as a sample vector. Here, K means to be generated by the dot product of all the training samples in the implicit feature space, which is similar to (8). The procedure of computing KNFST can be summarized as follows [19]:

- Step 1. Compute the kernel matrix K using (8).
- Step 2. Compute the orthogonal basis $(\tau_1, \dots, \tau_{n-c})$ from the set of **centered** vectors $\{K_{.1} - \bar{k}, K_{.2} - \bar{k}, \dots, K_{.n} - \bar{k}\}$ where $\bar{k} = \frac{1}{n} \sum_{i=1}^n K_{.i}$.
- Step 3. Set $P = (\tau_1, \dots, \tau_{n-c})$ and compute $S = P^T K (I - L) P$ where I is the $n \times n$ identity matrix, and $L = (L_l)_{l=1, \dots, c}$ is a $n \times n$ block diagonal matrix with entries $L_{ij} = 1/n_j$ if sample i belongs to class j . Solving $S\eta = 0$ yields vectors $\eta_1, \dots, \eta_{c-1}$.
- Step 4. KNFST is then given by AQ , where $A = (\eta_1, \dots, \eta_{c-1})$.

Ali and Chaudhuri [3] showed that the singularities of different classes are not necessarily distinct, losing their class identities, which is a major limitation of NFST-based classifiers.

B. Comparison between CPAI and NFST

Ali and Chaudhuri [3] showed that the singularities of different classes are not necessarily distinct, losing their class identities, which is a major limitation of NFST-based classifiers.

1. *Addressing the singularity problem:* NFST relies on the constraints $\varphi^T S_w \varphi = 0$ and $\varphi^T S_b \varphi > 0$. However, it offers no explicit control over the final locations in the projection space. CPAI directly addresses this by allowing the user to predefine distinct target projection values (b_j) for each class. The algorithm then explicitly constructs discriminant vectors to map samples towards these separated coordinates by solving $(K + \lambda I)\alpha = \beta$.
2. *Handling data dimensionality:* Linear NFST fundamentally requires linear independence of sample vectors, which generally necessitates $n \leq d$. **Kernel CPAI** overcomes this limitation by working with the $n \times n$ kernel matrix K . As long as $\lambda > 0$, a unique solution α exists regardless of the dimension's assumption.
3. *Implementation and Kernel choice:* The effectiveness of Kernel CPAI relies on selecting a suitable kernel function κ that results in a well-conditioned (or regularized) and invertible kernel matrix K , and captures the underlying data structure. Our experiments demonstrate that appropriate kernel choices (like Laplacian, RBF, Abel in our tests) significantly enhance performance over the linear case and potentially problematic kernels like Sigmoid. This highlights the importance of kernel selection, a common aspect of kernel methods.
4. **Spectral Behavior:** Unlike NFST's rigid binary selection of directions, CPAI performs continuous spectral weighting. By utilizing the scalar S_j and the Tikhonov regularization term (λ), CPAI ensures that global invariant features are preserved while singular amplification of noise is suppressed.

CRediT authorship contribution statement

Van-Hoi Nguyen: Conceptualization, Methodology, Validation, Formal analysis, Writing - Original Draft. **Xuan-Ha Nguyen:** Methodology, Software, Data curation, Investigation, Writing - Original Draft. **Kim-Hung Le:** Conceptualization, Methodology, Project administration, Writing - Review & Editing.